

7 **(a)** Sketch on a single Argand diagram

(i) the set of points for which $|z - 1 - 3i| = 3$, **[3]**

(ii) the set of points for which $\arg(z + 4) = \frac{1}{4}\pi$. **[3]**

(b) Find, in exact form, the two values of z for which $|z - 1 - 3i| = 3$ and $\arg(z + 4) = \frac{1}{4}\pi$. **[6]**